Topic 3: Trends in Interfaces (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

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01 INTERFACE TYPES

02 NATURAL USER INTERFACES AND BEYOND

03 WHICH INTERFACE?

2

INTERFACE TYPES

3

Overview

22 interface types covered

1. Command
2. Graphical
3. Multimedia
4. Virtual reality
5. Web
6. Mobile
7. Appliance
8. Voice
9. Pen
10. Touch
11. Touchless
12. Haptic
13. Multimodal
14. Shareable
15. Tangible
16. Augmented reality
17. Wearables
18. Robots and drones
19. Brain-computer
20. Smart
21. Shape-changing
22. Holographic

- Provide an overview of the diversity of interfaces
- Highlight the main design and research considerations for each of the interfaces

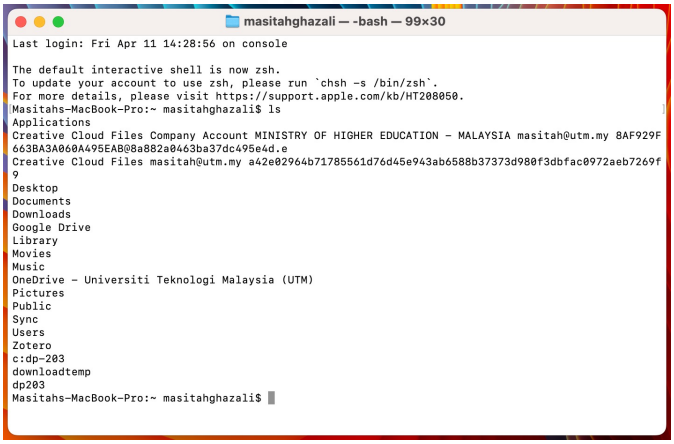
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1. Command Line Interfaces

- Commands such as abbreviations (for instance, **ls**) typed in at the prompt to which the system responds (for example, by listing current files)
- Some are hard wired at keyboard, while others can be assigned to keys
- Efficient, precise, and fast
- Large overhead to learning set of commands

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Command ls – lists what is on the laptop drive



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Research and Design Considerations

- Form, name types and structure are key research questions
- Consistency is most important design principle
  - For example, always use first letter of command
- Command interfaces popular for web scripting and shared document editing (e.g. Overleaf)

2. Graphical User Interfaces (GUIs)

- Xerox Star first WIMP gave rise to GUIs
- Windows
  - Sections of the screen that can be scrolled, stretched, overlapped, opened, closed, and moved around the screen using the mouse
- Icons
  - Pictograms that represent applications, objects, commands, and tools that were opened when clicked on
- Menus
  - Lists of options that can be scrolled through and selected
- Pointing device
  - A mouse controlling the cursor as a point of entry to the windows, menus, and icons on the screen

Example of First-Generation GUI

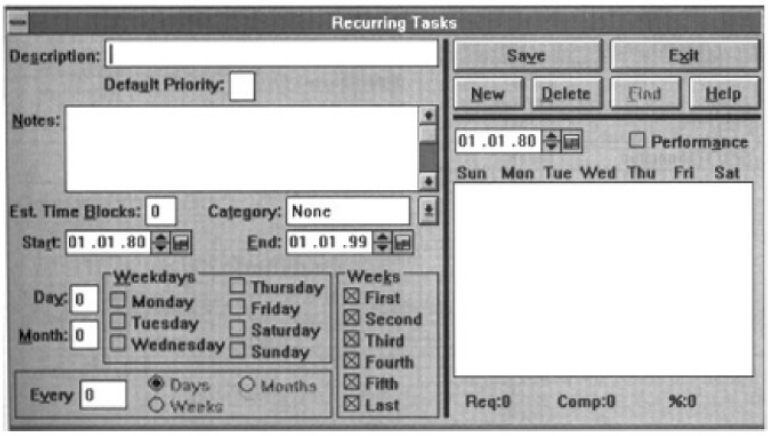
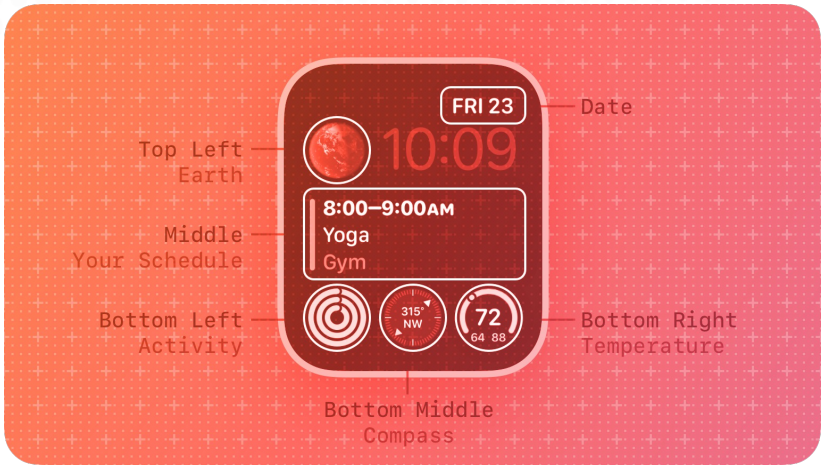


Figure 6.2 The boxy look of the first generation of GUIs. The window presents several check boxes, notes boxes, and options as square buttons

Source: Mullet, Kevin; Sano, Darrell, *Designing Visual Interfaces: Communication Oriented Techniques*, 1st, © 1995. Reproduced by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.

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Smartwatch Complication Display



An Apple watchOS complication display that include the features "circulars" (three kinds shown in the bottom line) and "inlines" (shown in the middle and the upper right-hand corner of the display)

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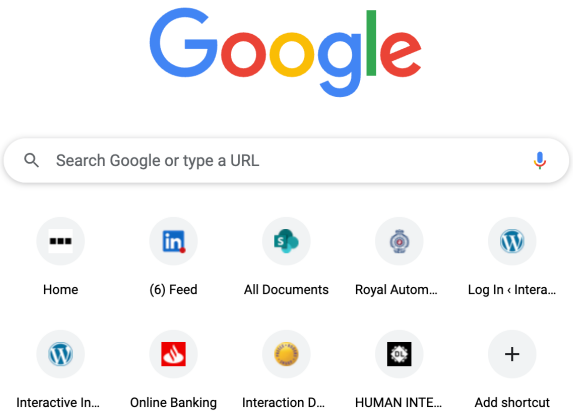
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Window Design

- Windows were invented to overcome the physical constraints of a computer display
- They enable more information to be viewed and tasks to be performed
- Scroll bars within windows enable more information to be viewed
- Multiple windows can make it difficult to find desired one
- Listing, tabbing, and thumbnails are techniques that can help

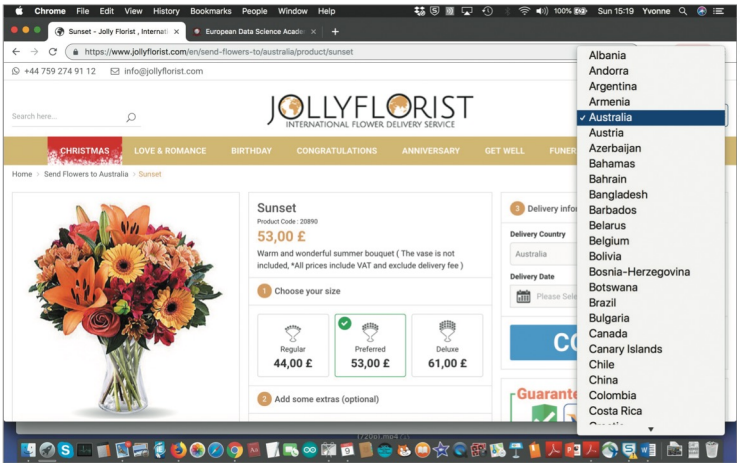
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Window Design: favicons of top sites visited below the Google search bar



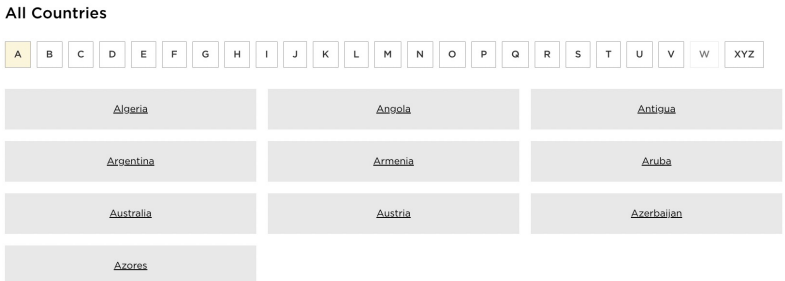
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Selecting a country from a scrolling window



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Is this method better? If so, why?



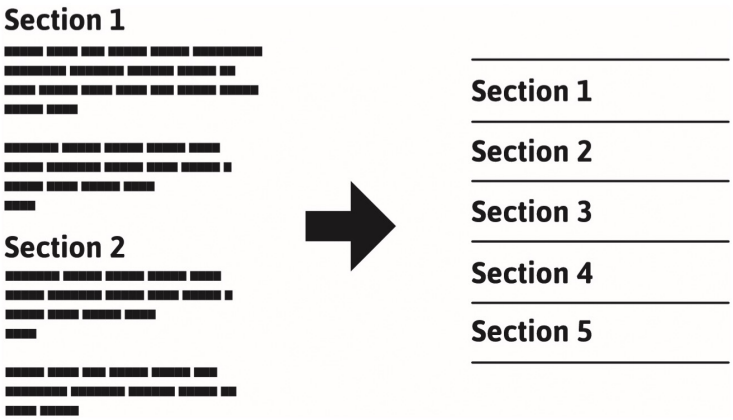
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Menu Styles

- **Flat list:** Good for showing large number of options at the same time when display is small
- **Drop down:** Shows more options on same screen (for example, cascading)
- **Pop-up:** When pressed, command key for relevant options
- **Contextual:** Provides access to often-used commands associated with a particular item
- **Collapsible:** Toggles between + and - icons on a header to expand or contract its contents
- **Mega:** All options shown using 2D drop-down layout

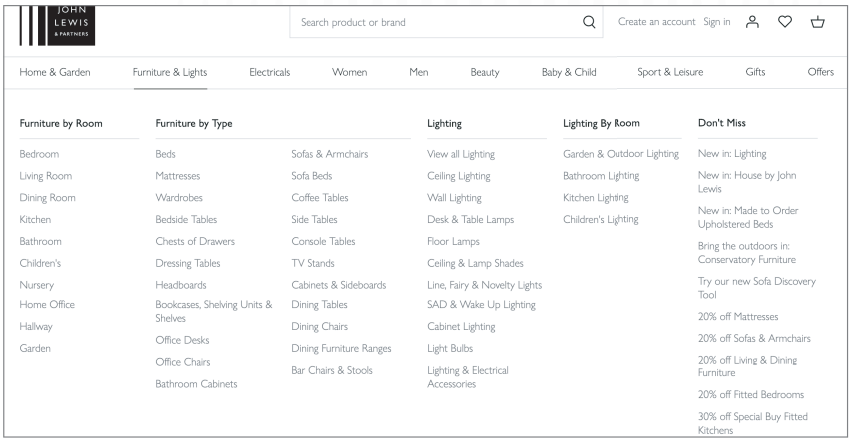
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Template for a Collapsible Menu



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A Mega Menu



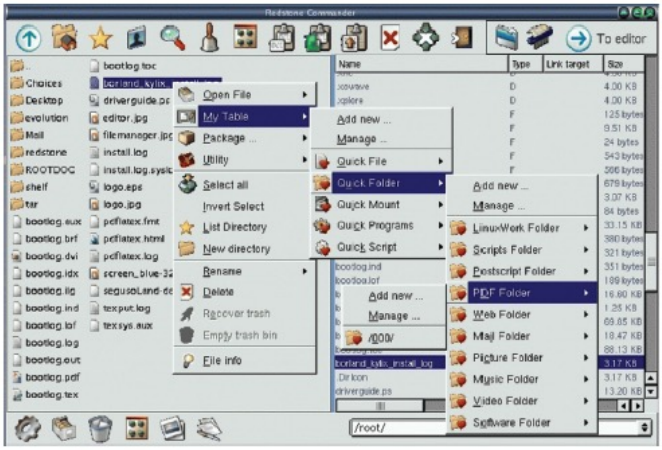
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Expanding Menus

- Enables more options to be shown on a single screen than is possible with a single flat menu
- More flexible navigation, allowing for selection of options to be done in the same window
- Most popular are cascading ones
  - primary, secondary and even tertiary menus
  - downside is that they require precise mouse control
  - can result in overshooting or selecting wrong options

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A Cascading Menu



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Contextual Menus

- Provide access to often-used commands that make sense in the context of a current task
- Appear when the user presses the Control key while clicking on an interface element
  - e.g., clicking on a photo in a website together with holding down the Control key results in options 'open it in a new window,' 'save it,' or 'copy it'
- Helps overcome some of the navigation problems associated with cascading menus

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Windows Jump List Menu

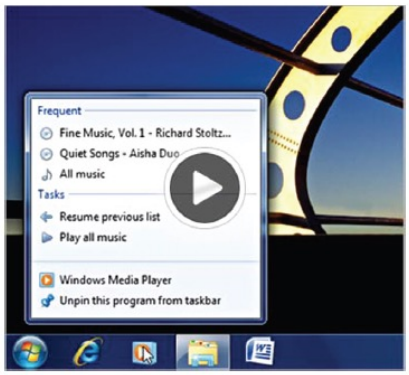


Figure 6.8 Windows jump list

Source: http://windows.microsoft.com/en-US/windows7/products/features/jump-lists.

Research and Design Considerations

- Window management
  - Enables users to move fluidly between different windows (and monitors)
- How to switch attention between windows without getting distracted
- Design principles of spacing, grouping, and simplicity should be used
- Which terms to use for menu options (for example, "front" versus "bring to front"
- Mega menus easier to navigate than drop-down ones

Icon Design

- Icons are assumed to be easier to learn and remember than commands
- Icons can be designed to be compact and variably positioned on a screen
- Now pervasive in every interface
  - For example, they represent desktop objects, tools (for example, a paintbrush), applications (for instance, a web browser), and operations (such as cut, paste, next, accept, and change)

Icons

- Since the Xerox Star days, icons have changed in their look and feel:
  - black and white → Color, shadowing, photorealistic images, 3D rendering, animation, flat
- Many designed to be very detailed and animated making them both visually attractive and informative
- Can be highly inviting, emotionally appealing, and feel alive

Icon Forms

- The mapping between the representation and underlying referent can be:
  - Similar (for example, a picture of a file to represent the object file)
  - Analogical (for instance, a picture of a pair of scissors to represent 'cut')
  - Arbitrary (such as the use of an X to represent 'delete')

- The most effective icons are similar ones
- Many operations are actions making it more difficult to represent them
  - Use a combination of objects and symbols that capture the salient part of an action

2 types of Apple Aqua icon styles



Flat 2D icons for a smartphone and a smartwatch



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Research and Design Considerations

- There is a wealth of resources for creating icons
  - Guidelines, style guides, icon builders, libraries, online tutorials
- Text labels can be used alongside icons to help identification for small icon sets
- For large icon sets (for instance, photo editing or word processing) can use the hover function

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3. Multimedia

- Combines different media within a single interface with various forms of interactivity
  - Graphics, text, video, sound, and animation
- Users click on links in an image or text
  - Another part of the program
  - An animation or a video clip is played
  - Users can return to where they were or move on to another place
- Can provide better ways of presenting information than a single media can

Pros and Cons

- Facilitates rapid access to multiple representations of information
- Can provide better ways of presenting information than can any media alone
- Can enable easier learning, better understanding, more engagement, and more pleasure
- Can encourage users to explore different parts of a game or story
- Tendency to play video clips and animations while skimming through accompanying text or diagrams

Multimedia learning app designed for tablet



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Research and Design Considerations

- How to design multimedia to help users explore, keep track of, and integrate the multiple representations
  - Provide hands-on interactivities and simulations that the user has to complete to solve a task
  - Provide quizzes, electronic notebooks, and games

- Multimedia good for supporting certain activities, such as browsing, but less optimal for reading at length

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4. Virtual Reality

- Computer-generated graphical simulations providing:
  - "the illusion of participation in a synthetic environment rather than external observation of such an environment" (Gigante, 1993)
- Provide new kinds of experience, enabling users to interact with objects and navigate in 3D space
- Create highly-engaging user experiences

Pros and Cons

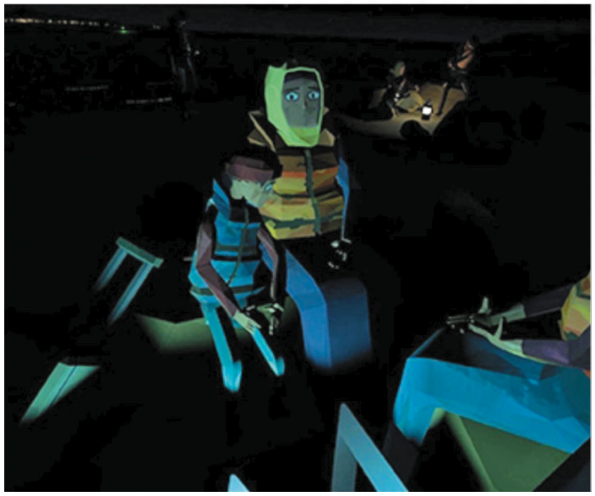
- Can have a higher level of fidelity with objects that they represent compared to multimedia
- Induces a sense of presence where someone is totally engrossed by the experience
  - "a state of consciousness, the (psychological) sense of being in the virtual environment" (Slater and Wilbur, 1999)
- Provides different viewpoints: first and third person
- Early head-mounted displays were uncomfortable to wear and could cause motion sickness and disorientation
- Lighter VR headsets are now available (for example, HTC Vive) with more accurate head tracking

Examples

- Video games
- Arcade games for social groups
- Well being and therapy for fears
- Experience how others feel emotions
  - For example, empathy and compassion
- Enrich user's planning experience for travel destinations
- Architecture, design, and education

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Polygon graphics used to represent avatars for the We Wait VR experience



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UTM Classroom of the Future



Offshore experience
Source: https://magicx.my/classroom-of-the-future/

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Research and Design Considerations

- Much research on how to design safe and realistic VRs to facilitate training
  - For example, flying simulators
  - Help people overcome phobias (for example, spiders or talking in public)
- Design issues
  - How best to navigate through them (for instance, first versus third person)
  - How to control interactions and movements (for example, by using head and body movements)
  - How best to interact with information (for instance by using keypads, controllers, pointing, and joystick buttons)
  - Level of realism to aim for to engender a sense of presence

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5. Web

- Early websites were largely text-based, providing hyperlinks
- Focus was on how best to structure information to enable users to navigate and access them easily and quickly
- Nowadays, more emphasis is on making pages distinctive, striking, and aesthetically pleasing
- Need to think of how to design information for different platforms—keyboard or touch?
  - For example, smartphones, tablets, and PCs

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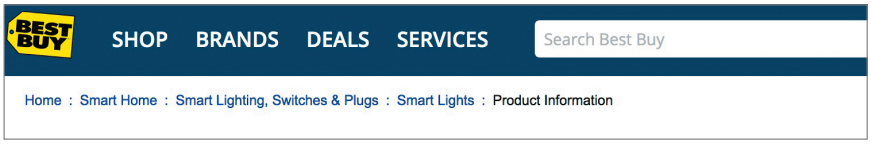
Usability versus Aesthetics

- Vanilla or multi-flavour design?
  - Ease of finding something versus aesthetic and enjoyable experience
- Web designers are:
  - "thinking great literature"
- Users read the web like a:
  - "billboard going by at 60 miles an hour" (Krug, 2014)
- Need to determine how to brand a web page to catch and keep 'eyeballs'

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Breadcrumbs for Navigation

- Breadcrumbs are category labels:
  - Enable users to look at other pages without losing track of where they have come from
  - Very usable
  - Enable one-click access to higher site levels
  - Attract first time visitors to continue to browse a website having viewed the landing page



Web Design Styles

- Responsive website design
  - the browser automatically resizes the layout, and changes the graphic design, fonts, and appearance depending on the screen size (smartphone, tablet, or PC) on which it is being displayed.
  - downsizing the content in this way makes it more time-consuming as more pages need to be loaded.
  - can also make it more fiddly navigating multiple pages and menus
- Infinite scrolling
  - websites are designed to enable browsing content on one long page
  - avoids a visitor needing to wait for pages to load when clicking on them
  - navigation is largely done by swiping across or down the page until the end is reached
  - a side effect is the tendency to glance while scrolling without focusing on individual items

In-your-face Web Ads

- Web advertising is often intrusive and pervasive
- Flashing, aggressive, persistent, and annoying
- Often requires action to get rid of
- What is the alternative?
  - Use of ad blockers

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Research and Design Considerations

- Many books and guidelines on website design
- Veen's (2001) three core questions to consider when designing any website:
  - Where am I?
  - Where can I go?
  - What's here?

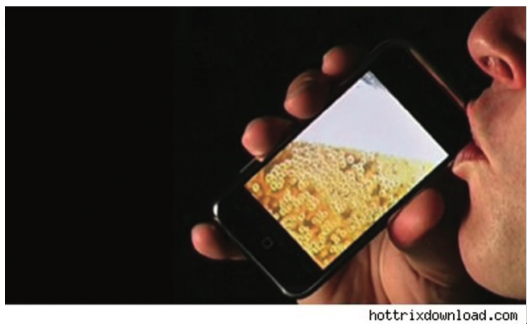
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6. Mobile

- Handheld devices intended to be used while on the move
- Have become pervasive, increasingly used in all aspects of everyday and working life
  - For example, phones, fitness trackers, and smartwatches
- Larger-sized tablets used in mobile settings
  - Including those used by flight attendants, marketing professionals, and at car rental returns

The Advent of the iPhone App

- A whole new user experience that was designed primarily for people to enjoy
  - many apps not designed for any need, want or use but purely for idle moments to have some fun
  - e.g. iBeer developed by magician Steve Sheraton
  - ingenious use of the accelerometer that is inside the phone



QR Codes and Smartphones





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Mobile Challenges

- Smaller screens, small number of physical keys and restricted number of controls
- Innovative physical designs including:
  - roller wheels, rocker dials, up/down 'lips' on the face of phones, 2-way and 4-way directional keypads, softkeys, silk-screened buttons
- Usability and preference varies
  - depends on the dexterity and commitment of the user
- Smartphones overcome mobile physical constraints through using multi-touch displays

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Research and Design Considerations

- Mobile interfaces can be cumbersome to use for those with poor manual dexterity or 'fat' fingers
- Key concern is hit area:
  - Area on the phone display that the user touches to make something happen, such as a key, an icon, a button, or an app
  - Space needs to be big enough for all fingers to press accurately
  - If too small, the user may accidentally press the wrong key
  - Fitts' law can be used to help design right spacing
    - Minimum tappable areas should be 44 points x 44 points for all controls

7. Appliances

- Everyday devices in home, public places, or car
  - For example, washing machines, remotes, toasters, printers, and navigation systems)
- And personal devices
  - For instance, digital clock and digital camera
- Used for short periods
  - For example, starting the washing machine, watching a program, buying a ticket, changing the time, or taking a snapshot
- Need to be usable with minimal, if any, learning

Simple Toaster Control



Research and Design Considerations

- Need to design as transient interfaces with short interactions
- Simple interfaces
- Consider trade-off between soft and hard controls
  - e.g. buttons or keys, dials or scrolling
- Understand the innate human ability to manipulate physicality (Ghazali and Dix)
  - The relationship between the physical controllers with the digital underlying functionalities

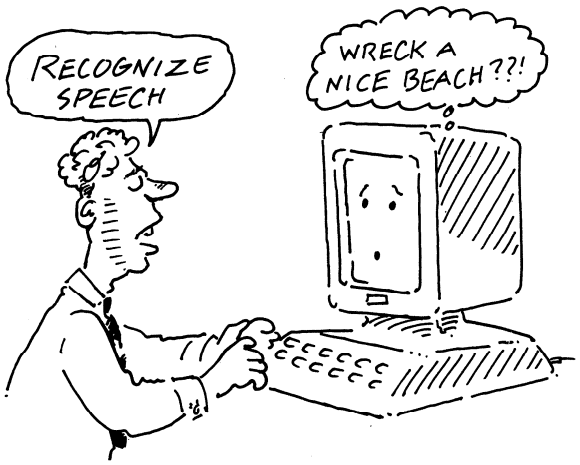
8. Voice

- Involves a person talking with a spoken language app, for example, timetable, travel planner, or phone service or chatbots
- Used mainly for inquiring about specific information, for example, flight times or to perform a transaction, such as buying a ticket
- Mostly a voice system is reactive responding to a person's queries
- Also used by people with visual impairments
  - For example, speech recognition word processors, page scanners, web readers, and home control systems

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Have speech interfaces come of age?



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Modelling Human Conversations

- People often interrupt each other in a conversation
  - Especially when ordering in a restaurant, rather than let the waiter go through all of the options
- Speech technology has a similar feature called 'barge-in'
  - Users can choose an option before the system has finished listing all of the options available



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Structuring Voice Dialogs

- Directed dialogs are where the system is in control of the conversation
  - Where it asks specific questions and requires specific responses
- More flexible systems allow the user to take the initiative:
  - For example, "I'd like to go to Paris next Monday for two weeks."
- Guided prompts can help callers back on track
  - For example, "Sorry I did not get all that. Did you say you wanted to fly next Monday?"



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Voice Assistants (for e.g. Alexa)

- Allow all to use rather than being single use
- Support families playing games, interactive storytelling, jokes, and so forth
- Can encourage social and emotional bonding
- Young children (under 4), however, find it difficult to be understood by the voice assistants
  - Frustrating for them

Research and Design Considerations

- How to design systems that can keep conversation on track
  - Help people navigate efficiently through a menu system
  - Enable them to recover easily from errors
  - Guide those who are vague or ambiguous in their requests for information or services
- Type of voice actor (for example, male, female, neutral, or dialect)
  - Do people prefer to listen to and are more patient with a female or male voice, a northern or southern accent?

9. Pen-based Devices

- Enable people to write, draw, select, and move objects at an interface using light pens or styluses
  - Capitalise on the well-honed drawing skills developed from childhood
- Smartpens use a combination of ordinary ink pen with digital camera that digitally records everything written with the pen on special paper

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The LiveScribe Echo 2 Smartpen



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Pros and Cons

- Allows people to annotate existing documents quickly and easily
- Can be used to fill in paper-based forms that can readily be converted to a digital record using standard typeface
- Can be used by remote teams to communicate and work on the same documents

- Can be difficult to see options on the screen because a user's hand can occlude part of it when writing
- Can have lag and feel clunky

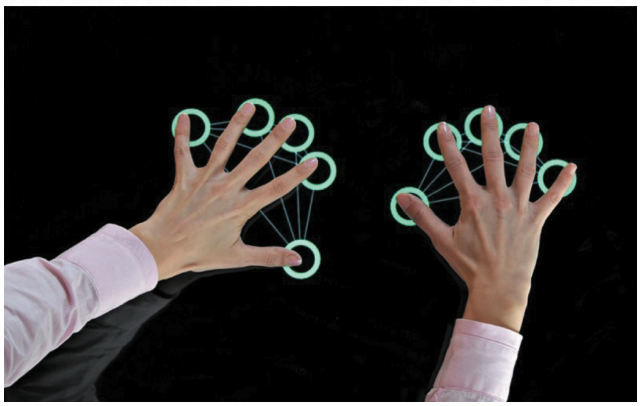
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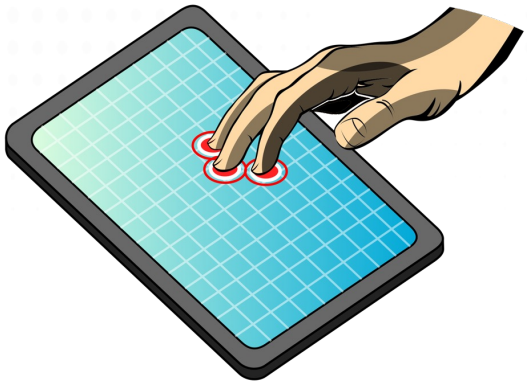
10. Touchscreens

- Single touchscreens are used in walk-up kiosks (such as ticket machines and ATMs) to detect the presence and location of a person's touch on the display
- Multi-touch surfaces support a range of more dynamic finger tip actions, for example, swiping, flicking, pinching, pushing, and tapping
- They do so by registering touches at multiple locations using a grid
- Now used for many kinds of displays, such as smartphones, iPods, tablets, and tabletops
  - Supports one and two hand gestures, including tapping, zooming, stretching, flicking, dwelling, and dragging

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A Multi-touch Surface





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Research and Design Considerations

- Provides fluid and direct styles of interaction involving freehand and pen-based gestures for certain tasks
- Core design concerns include whether size, orientation, and shape of touch displays effect collaboration
- Much faster to scroll through wheels, carousels, and bars of thumbnail images or lists of options by finger flicking
- Gestures need to be learned for multi-touch, so a small set of gestures for common commands is preferable
- More cumbersome, error-prone, and slower to type using a virtual keyboard on a touch display than using a physical keyboard

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11. Touchless

- Gestures involve moving arms and hands to communicate
- Uses camera recognition, sensor, and computer vision techniques
  - Recognise people's arm and hand gestures in a room
  - Gestures need to be presented sequentially to be understood (compare with the way sentences are constructed)

Touchless interface in the operating theater



Recognises core gestures for manipulating MRI or CT images using Microsoft Kinect

Entertainment (public space)



MaGICX Johor Port 2015
Source: https://magicx.my/

Car Dashboards using Gestures

- BMW's iDrive enables drivers to control certain functions with the use of hand gestures captured by a 3D camera
  - e.g. turning up or down the audio volume, For audio control these comprise of rotating the index finger clockwise (up) or anti-clockwise (down)
- Gesture control sensors are also positioned in the roof lining of the car
- To ensure they are safe, the gestures were developed and tested in the simulator

Research and Design Considerations

- How does computer recognize and delineate user's gestures?
  - Start and end points?
  - Difference between deictic and hand waving
- How realistic must the mirrored graphical representation of the user be in order for them to be believable?

12. Haptic

- Provide tactile feedback
  - By applying vibration and forces to a person's body, using actuators that are embedded in their clothing or a device they are carrying, such as a smartphone
- Vibrotactile feedback can be used to simulate the sense of touch between remote people who want to communicate
- Ultrahaptics creates the illusion of touch in midair using ultrasound to make the illusion of 3D shapes

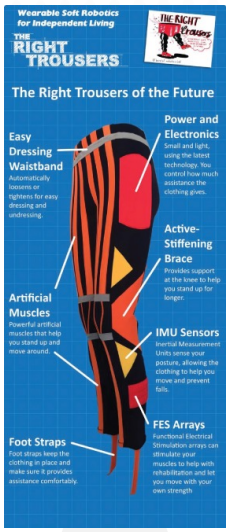
Realtime Vibrotactile Feedback

- Provides nudges when playing violin incorrectly
- Uses motion capture to sense arm movements that deviate from model
- Nudges are short vibrations on arms and hands



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Exoskeleton with artificial muscles that uses bubble haptic feedback



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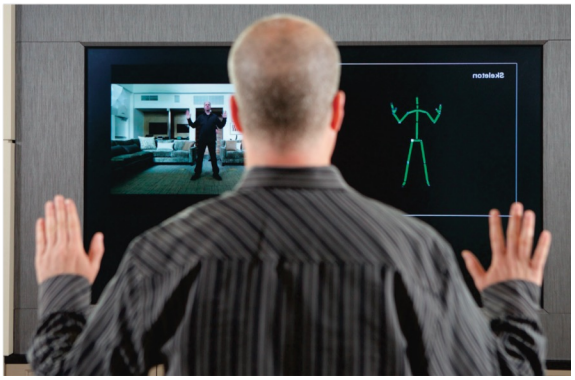
Research and Design Considerations

- Where best to place actuators on body
- Whether to use single or sequence of 'touches'
- When to buzz and how intense
- How does the wearer feel it in different contexts?
- What kind of new smartphone/smartwatch apps can use vibrotactile creatively?
  - For example, slow tapping to feel like water drops meant to indicate that it is about to rain, and heavy tapping to indicate a thunderstorm is looming

13. Multimodal Interfaces

- Provide enriched user experiences
  - By multiplying how information is experienced and detected using different modalities, such as touch, sight, sound, and speech
  - Support more flexible, efficient, and expressive means of human-computer interaction
  - Most common combination is speech and vision
- Can be combined with multi-sensor input to enable other aspects of the human body to be tracked
  - For example, eye gaze, facial expression, and lip movements
  - Provides input for customising user interfaces

Tracking a person's movement



- Kinect camera can detect multimodal input in real time using RGA camera for facial recognition and gestures, depth camera for movement tracking, and microphones for voice recognition
- Used to build model of person and represented as avatar on display programmed to move just like them

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Sending modalities via smartphone



Send taste, smell, touch via smartphone: https://learningenglish.voanews.com/a/smartphones-may-be-able-to-send-tastes-smeel-touches/2634824.html

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Research and Design Considerations

- Need to recognise and analyse user behavior, for example, speech, gesture, handwriting, or eye gaze
- Much harder to calibrate these than single modality systems
- What is gained from combining different input and outputs
- Is talking and gesturing, as humans do with other humans, a natural way of interacting with a computer?

14. Shareable Interfaces

- Designed for more than one person to use:
  - Provide multiple inputs and sometimes allow simultaneous input by co-located groups
  - Large wall displays where people use their own pens or gestures
  - Interactive tabletops where small groups interact with information using their fingertips
    - For example, DiamondTouch, Smart Table, and Surface

A smartboard and an interactive tabletop interface



(a)



(b)

Benefits

- Provide a large interactional space that can support flexible group working
- Can be used by multiple users
  - Can point to and touch information being displayed
  - Simultaneously view the interactions and have the same shared point of reference as others
- Can support more equitable participation compared with groups using single PC

Research and Design Considerations

- Core design concerns include whether size, orientation, and shape of the display have an effect on collaboration
- Horizontal surfaces compared with vertical ones support more turn-taking and collaborative working in co-located groups
- Providing larger-sized tabletops does not improve group working but encourages more division of labor
- Having both personal and shared spaces enables groups to work on their own and in a group
- A design challenge is how to bridge across devices, applications and time such that the peoples' interactions are seamless

15. Tangible Interfaces

- Type of sensor-based interaction, where physical objects, for example, bricks, are coupled with digital representations
- When a person manipulates the physical object/s, it causes a digital effect to occur, for example, an animation
- Digital effects can take place in a number of media and places, or they can be embedded in the physical object

Examples

- Flow Blocks
  - Depict changing numbers and lights embedded in the blocks
  - Vary depending on how they are connected together
- Urp
  - Physical models of buildings moved around on tabletop
  - Used in combination with tokens for wind and shadows → Digital shadows surrounding them to change over time
- MagicCubes
  - Connect physical electronic components and sensors to make digital events occur (for example, change color depending on how much shaken)

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Learning to code and create with the tangible MagicCubes



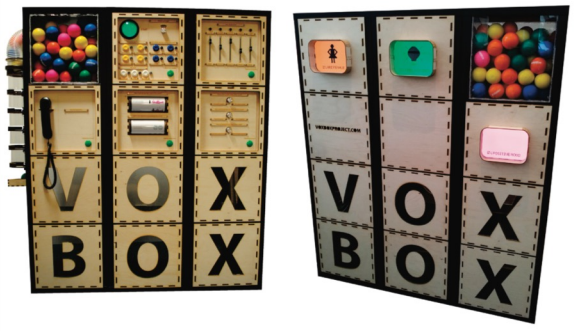
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Benefits

- Can be held in one or both hands and combined and manipulated in ways not possible using other interfaces
  - Allows for more than one person to explore the interface together
  - Objects can be placed on top of each other, beside each other, and inside each other
  - Encourages different ways of representing and exploring a problem space
- People are able to see and understand situations differently
  - Can lead to greater insight, learning, and problem-solving than with other kinds of interfaces
  - Can facilitate creativity and reflection

VoxBox

A tangible system that gathers opinions at events through playful and engaging interaction (Goldsteijn et al., 2015)



Research and Design Considerations

- What kinds of conceptual frameworks to use to help identify novel and specific features
- What kind of coupling to use between the physical action and digital effect
  - If it is to support learning, then an explicit mapping between action and effect is critical
  - If it is for entertainment, then it can be better to design it to be more implicit and unexpected
- What kind of physical artifact to use
  - Bricks, cubes, and other component sets are most commonly used because of flexibility and simplicity
  - Stickies and cardboard tokens can also be used for placing material onto a surface
- With what kinds of digital outputs should tangible interfaces be combined?

16. Augmented Reality

- Digital representations are superimposed on physical objects or the environment
- Pokémon Go made it a household game
  - Used smartphone camera and GPS to place virtual characters onto objects in the environment as if they really are there
- Many other applications including medicine, navigation, air traffic control, games, and everyday exploring

Other Examples

- In medicine
  - Virtual objects, for example, x-rays and scans, are overlaid on part of a patient's body
  - Aid the physician's understanding of what is being examined or operated

- In air traffic control
  - Dynamic information about aircraft overlaid on a video screen showing the real planes, and so on landing, taking off, and taxiing
  - Helps identify planes difficult to make out

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Augmented reality overlay on a car windshield



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Devices for Viewing AR

- AR can be viewed through headsets, smartphones and glasses (e.g., SnapChat's AR spectacles).
- Headsets can provide a more immersive experience, but the downside is they are cumbersome to wear and fiddly to calibrate
- Convenient and easy-to-wear AR glasses are beginning to appear

Mobile AR BoboiBoy



AR that uses forward facing camera

- Enables virtual try-ons (for example, Snapchat filters, Zoom filters)
- AT mirrors set up in retail stores for trying on make-up, sunglasses, jewelry
  - Convenient, engaging, and easy to compare more choices
  - But cannot feel the weight, texture, or smell of what is being tried on
- Can be used to enable users to step into a character (for example, David Bowie, Queen Victoria)

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Singers trying on the virtual look of two characters from the opera Akhnaten



(a) (b)

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Research and Design Considerations

- What kind of digital augmentation?
  - When and where in physical environment?
  - Needs to stand out but not distract from ongoing task
  - Needs to be able to align with real world objects
  - What happens if the AR is slightly off?
- What kind of device?
  - Smartphone, tablet, head up display or other?

17. Wearables

- First developments were head- and eyewear-mounted cameras that enabled user to record what was seen and to access digital information
- Since then, jewelry, head-mounted caps, smart fabrics, glasses, shoes, and jackets have all been used
  - Provides the user with a means of interacting with digital information while on the move
- Applications include automatic diaries, tour guides, cycle indicators, and fashion clothing

Google Glass (2014)



Why was there so much excitement and concern about people filming what they could see right in front of them?

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Other Examples









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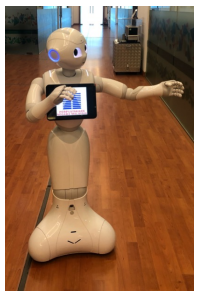
Research and Design Considerations

- Comfort
  - Needs to be light, small, not get in the way, fashionable, and preferably hidden in the clothing
- Hygiene
  - Is it possible to wash or clean the clothing once worn?
- Ease of wear
  - How easy is it to remove the electronic gadgetry and replace it?
- Usability
  - How does the user control the devices that are embedded in the clothing?

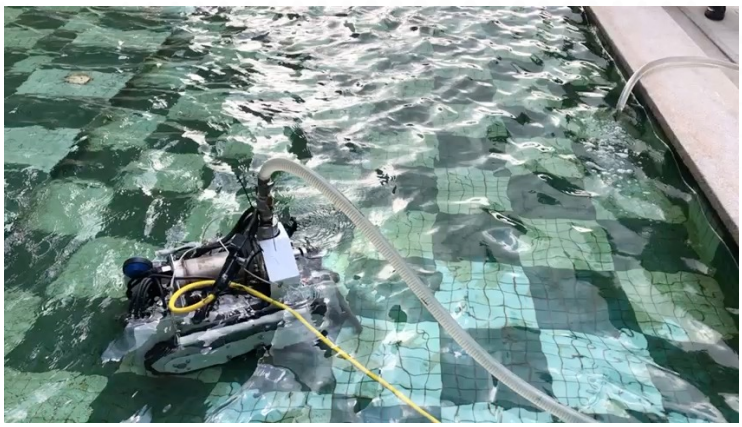
18. Robots …

- Main types
  - Remote robots used in hazardous settings
    - Can be controlled to investigate bombs and other dangerous materials
  - Domestic robots helping around the house
    - Can pick up objects and do daily chores like vacuuming
  - Pet robots as human companions
    - Have therapeutic qualities, helping to reduce stress and loneliness
  - Sociable robots that work collaboratively with humans
    - Encourage social behaviors





Cleaning Robot



UTM CAIRO: ROV Pleco to clean tank, pool

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Social Robots: Mel and Paro

- Cute and cuddly
- Can open and close eyes and make sounds and movements





Source: Images courtesy of Mitsubishi Electric Research Labs.

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18. … and Drones

- Unmanned aircraft that are controlled remotely and used in a number of contexts
  - Medical supplies, groceries and carrying drinks and food to people at festivals and parties
  - Construction and agricultural applications, such as flying them over vineyards and fields to collect data about crops
  - Firework displays
  - Helping to track poachers in wildlife parks in Africa
- Can fly low and and stream photos to a ground station where images can be stitched together into maps
- Can be used to determine the health of a crop, or when it is the best time to harvest the crop

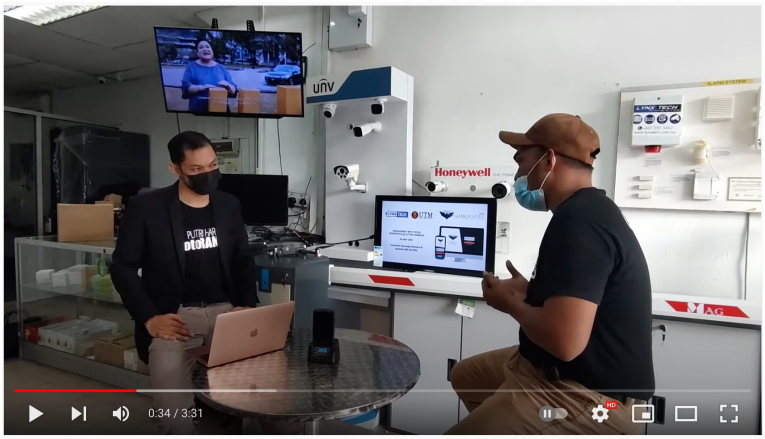
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Drone being used to survey the state of a vineyard



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Drone to patrol surrounding



GuardExpert Pro, an Internet-Of-Thing (IoT) based security guard patrolling system. GUARD EXPERT PRO

Research and Design Considerations

- How do humans react to physical robots designed to exhibit behaviors (for example, making facial expressions) compared with virtual ones?
- Should robots be designed to be human-like or look like and behave like robots that serve a clearly-defined purpose?
- Should the interaction be designed to enable people to interact with the robot as if it was another human being or more human-computer-like (for example, pressing buttons to issue commands)?
- Is it acceptable to use unmanned drones to take a series of images or videos of fields, towns, and private property without permission or people knowing what is happening?

19. Brain-Computer Interfaces

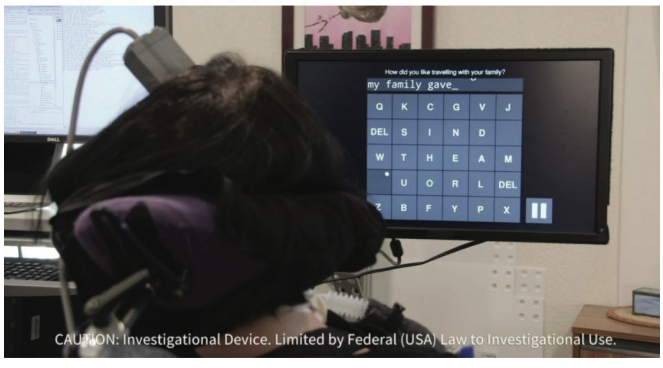
- Brain-computer interfaces (BCI) provide a communication pathway between a person's brain waves and an external device, such as a cursor on a screen
- Person is trained to concentrate on the task, for example, moving the cursor
- BCIs work through detecting changes in the neural functioning in the brain
- BCIs apps:
  - Games (for example, Brain Ball)
  - Enable people who are paralysed to control robots

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Examples



Brainball



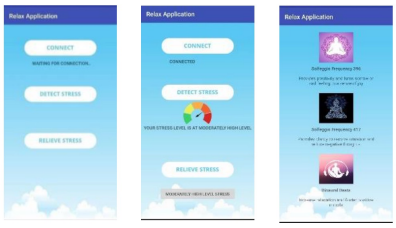
A brain-computer interface being used by a woman who is paralyzed to select letters on the screen

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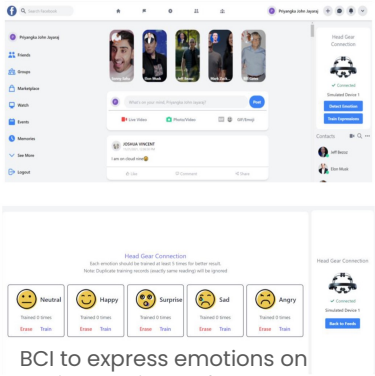
Other Examples



Using BCI to control and manipulate different types of mobile applications
Source: N.A.Iahad, A.B.G. Ahmed-Gaber, M. Ghazali. 2018. A Conceptual Model for Mobile Interaction using Brain Computer Interface. International Journal of Future Generation Communication & Networking (IJFGCN)



Mobile BCI to reduce stress among students
Source: P.J. Jayaraj, M. Ghazali and A.Gaber. Relax: Mobile Brain-Computer Interface App to Reduce Stress among Students. In Special Proceedings of 5th Asian CHI Symposium 2021.



BCI to express emotions on social media platform
Source: P. J. Jayaraj and M. Ghazali. User's Perspectives on Expressing Emotions on Social Media Platform with mobile Brain-Computer Interface. In Proceedings of 3rd Fusion Symposium 2021

Research and Design Considerations

- What is the best way to lay out letters on a digital screen so that the target space for each letter is sufficiently wide enough to allow for it to be selected using BCI?
- Ethical concerns are raised by brain-computer interfaces that are being developed to work out what someone is thinking
  - Our thoughts have always been private and making them interpretable by machines, which in turn could be accessed and read by other people, raises privacy concerns
- How accurate is BCI in what it infers to be someone's thoughts?

20. Smart Interfaces

- Smart: phones, speakers, watches, cars, buildings, cites
- Smart refers to having some intelligence and connected to the internet and other devices
- Context-aware
  - Understand what is happening around them and execute appropriate actions, for example, a Nest thermostat
- Human-building interaction
  - Buildings are designed to sense and act on behalf of the inhabitants but also allow them to have some control and interaction with the automated systems

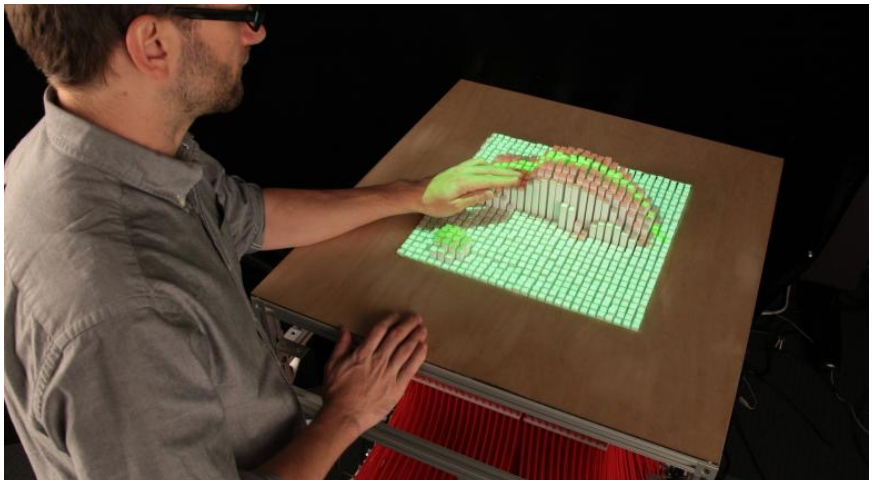
Research and Design Considerations

- Much current research is concerned with human values, needs, and priorities when addressing people's interactions with smart environments
- Do smart interfaces amplify our actions and remain attentive to our goals or are they becoming more autonomous?
- Do we need new metaphors to make understanding and using smart interfaces easier?
  - What might these be?
- Another approach is to imbue smart technologies, like speakers and robots, with personalities
  - Can promote user trust and acceptance

21. Shape Changing Interfaces

- Use physical shape change as input and output to systems
  - e.g. physical 3D bar chart that is positioned in a grid where a matrix of 3D rods move up and down to convey changes in a digital dataset
- These kinds of dynamic physical representations have been found to help people with visualisation tasks
- Shape-changing interfaces have also been developed as a form of dynamic material to explore novel interaction possibilities.
- Provide a different way of interacting with content compared with reading and touching digital displays

InFORM: a shape changing interface



Physicalisations

- Physical artefacts that are designed to encode data in specific materials
- An example is Physikit (Houben et al., 2016) which is a physical-digital system comprising a set of physical cubes that convey digital properties and are programmed to visualize real-time environmental data in the home
- A main benefit is to make data more accessible and enable people to more readily connect with the context in which the data is being collected or is changing over time (Sauvé et al, 2022)

Research and Design Considerations

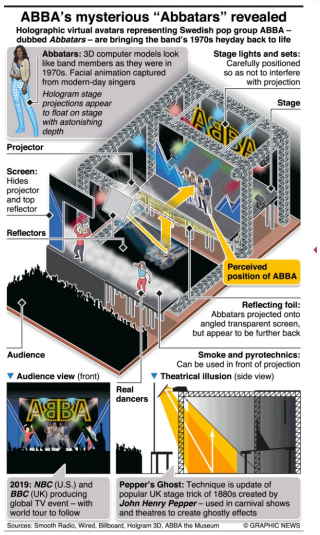
- Provide new opportunities for experiencing and exploring data using the sense of touch as well as sight
- Does this facilitate enhanced understanding and engagement with a dataset?
- Design considerations include what is the optimal size for a grid of physical rods and how many physicalisations to design in a set that people can learn and remember what they mean

22. Holographic Interfaces

- Create the illusion of a 3D person being present through using various forms of trickery
- Advances in projection and display technology have enabled these kinds of digital representations to appear quite convincing.
  - e.g. the Proto system lets people beam themselves to a remote location and interact with the people there
- The show Abba Voyage, first launched in 2022 feature avatars (dubbed 'ABBAtars'), depicting the group as they appeared in the late 70s
- The technology used includes a 65million pixel screen and in motion capture technology

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Examples



ABBA's holographic projections



UTM's Holo Professor

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Research and Design Considerations

- What is the best way to represent people in virtual spaces so that they feel comfortable, are engaging to interact with, feel natural and do not appear creepy?
- Design considerations include what size the holographs should be and how other's viewing them can interact and communicate with those being projected into their space

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NUI AND BEYOND

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From GUIs to Natural Interfaces

- Interfaces have shifted from screen-based to more embodied, intuitive forms. NUIs aim to reduce the cognitive load by leveraging skills we already use in the real world.
- Traditional interfaces from GUI → mobile → touch → wearables
- Emerging trends include holographic and smart interfaces
- Future direction requires more **natural** interactions
- Natural User Interfaces (NUIs):
  - Interact using voice, hands, gestures, facial expressions
  - Aim to replicate real-world human interaction

The Challenges with Naturalness

- Even 'natural' gestures can feel unnatural depending on context. Simplicity works best. More complex tasks may need a hybrid of natural and traditional inputs.
- NUIs should be intuitive—but context matters:
  - Is it easier to say 'open' or flick a switch?
  - Raising arms to change a TV channel vs pressing a button
- Don Norman's naturalness depends on:
  - Required learning
  - Task complexity
  - Need for speed & precision
- Example: sensor-based faucets work well with limited functions
  - But complex gestures for flow/temp? Less intuitive

Beyond Natural: Smart, Inclusive, Adaptive

- Interfaces of the future won't just be natural—they'll be smart, inclusive, and adaptive.
  - Gestural/voice interfaces enhance accessibility
    - e.g., iPhone VoiceOver for visually impaired
  - Brain-Computer Interfaces (BCIs): powerful but unnatural
    - Require significant training
  - Combining sensors (brain, body, environment) enables:
    - Real-time emotional/cognitive adaptation
    - Continuous & discrete inputs
- Merging sensor tech with AI opens up entirely new possibilities for human-computer collaboration.
    - AI empowers systems to collaborate with users

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WHICH INTERFACE?

....

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Which Interface?

- Which interface to use will depend on task, users, context, cost, robustness, and so on
- Is multimedia better than tangible interfaces for learning?
- Is speech as effective as a command-based interface?
- Is a multimodal interface more effective than a mono-modal interface?
- Will wearable interfaces be better than mobile interfaces for helping people to find information in foreign cities?
- Are virtual environments the ultimate interface for playing games?
- Are shareable interfaces better at supporting communication and collaboration compared with using networked desktop PCs?

Summary

- Many innovative interfaces have emerged in the last 30 years, including speech, wearable, mobile, brain, and tangible
- This raises many design and research questions as to decide which to use
  - For example, how best to represent information to the user so that they can carry out ongoing activity or task
- New smart interfaces that are context-aware and monitor people
- Raising new ethical issues concerned with what data is being collected and what it is used for